

Regression Methods to Analyze the Sales of a Souvenir Shop

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1 – Introduction

This project aims to gain a deeper understanding of time-series regression and nonparametric regression methods.

This data explores a list of monthly sales for a souvenir shop at a beach resort town located in Queensland, Australia from January 1987 – December 1993. This data was originally published by Wheelwright and Hyndman in 1998. This problem aims to use the sales data from 1987 to 1992 to predict the sales of 1993 using the exponentially weighted moving average (EWMA.) The prediction will then be analyzed against the actual 1993 sales from the data set.

Going solely off of the nature of the data, it is common sense to assume that there will be seasonality to this data. This data set holds a record of sales from a shop in a **beach resort**, which indicates that sales would be higher in the summer. There could also be several other factors at play, such as the amount of items in stock, the number of visitors to the beach, open hours, etc. However, we will assume these external factors are kept constant and the number of visits only increases due to increase in population.

1.1 – Data Description and Interpretation

Figure 1 shows the plot of sales from 1987 to 1994 in millions of people. As hypothesized previously, there is a seasonality to the data which peaks at the end of each year and drops back to a baseline towards the beginning of year. The usual visiting time of Australian beaches is December, so this trend makes sense. There are also several random fluctuations within the number of sales and an increasing baseline trend over time. The magnitude of fluctuations increases over time, with the highest values and most significant changes occurring in the last few years (1992-1994). However, to reduce variability, stabilize variance, linearize exponential trends and improve interpretability, we must perform a **logarithmic transformation** so we can do more analysis on the data.

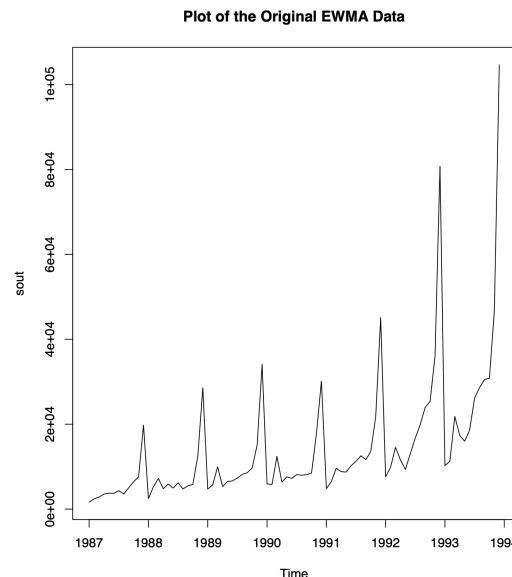


Figure 1: Original Plot of Monthly Sales Data

Figure 2 shows the natural log transformation of the original EWMA data. After this transformation, it is obvious to see that the random fluctuations and the seasonality effect are almost constant throughout every year.

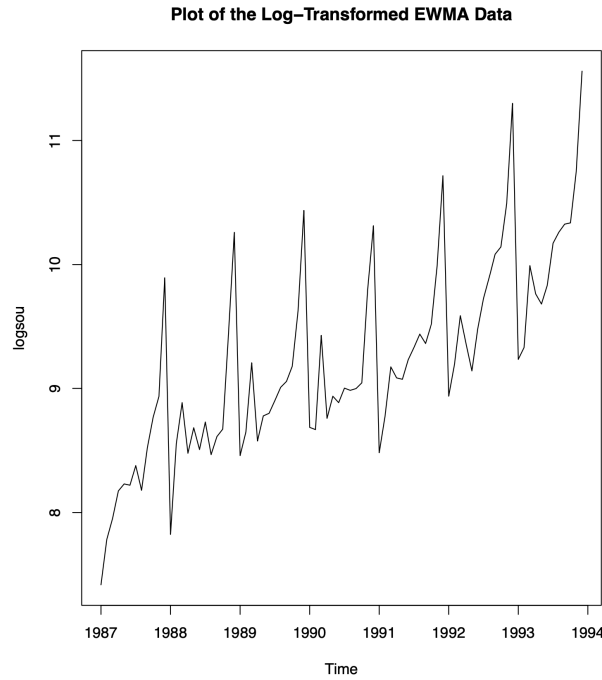


Figure 2: Log-Transformed EWMA Data

Now that we have successfully transformed our data using the logarithmic function, we can see that the variance appears more stable over the entire period. The spikes and troughs are more uniformly distributed, making the data easier to model and interpret. The upward trend is also more clearly visible and consistent throughout the period. The exponential growth trend in the original data has been linearized, making the overall growth pattern easier to analyze.

3 EWMA techniques are applied to predict the sales number in 1993. **Table 1** summarizes the methodology used to predict the model and the corresponding equations. The number on EWMA shows the amount of factors that the model holds. The three factors are: **level, trend, season**. The following equations represent one period ahead forecasts:

Method	Equation	Variables
Single EWMA	Forecasting Equation: $\hat{Y}_t = Y_t \alpha + (1 - \alpha) \hat{Y}_{t-1}$	$\hat{Y}_t \rightarrow$ forecasted value at time t $Y_t \rightarrow$ actual value at time t $\alpha \rightarrow$ level smoothing factor ($0 \leq \alpha \leq 1$)
Double EWMA	$L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + T_{t-1})$ $T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$ Forecasting Equation: $\hat{Y}_t = L_{t-1} + T_{t-1}$	$L_t \rightarrow$ estimated level at time t $T_t \rightarrow$ estimated trend at time t $\beta \rightarrow$ trend smoothing factor ($0 \leq \beta \leq 1$) $\alpha \rightarrow$ level smoothing factor ($0 \leq \alpha \leq 1$) $Y_t \rightarrow$ actual value at time t

Additive Triple EWMA	$L_t = \alpha(Y_t - s_{t-m}) + (1 - \alpha)(L_{t-1} + T_{t-1})$ $T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$ $S_t = \gamma(Y_t - L_t) + (1 - \gamma)s_{t-m}$ Forecasting Equation: $\hat{Y}_{t+h} = L_t + hT_t + S_{t-m+h}$	→ some variables in previous box S_t → estimate seasonal component at time t γ → seasonal smoothing factor ($0 \leq \gamma \leq 1$) m → number of periods in a season $\hat{Y}_{t+h}$ → forecast for h periods ahead
Multiplicative Triple EWMA	$L_t = \alpha\left(\frac{Y_t}{S_{t-m}}\right) + (1 - \alpha)(L_{t-1} + T_{t-1})$ $T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$ $S_t = \gamma\left(\frac{Y_t}{L_t}\right) + (1 - \gamma)s_{t-m}$ $\hat{Y}_{t+k} = L_t + hk + S_{t-m+k}$	→ some variables in previous box $\hat{Y}_{t+k}$ → forecasted value at time t + k.

Table 1: Different EWMA Methods for Prediction/Forecasting

The single EWMA has neither the trend or seasonal components, but contains the smoothing factor α . The double EWMA includes the trend factor β but not the seasonality factor γ . The triple EWMA includes the level factor α , trend factor β and seasonal factor γ .

2 – Model Building

To build the model that will predict the sales in 1993, we need to use the log transformed data and the Holt-Winters method in R. This function attempts to find the optimal values of $\alpha/\beta/\gamma$ by minimizing the squared on-step prediction error if they are NULL. For all EWMAs, we set $\alpha/\beta/\gamma$ to FALSE. We will then use the new fitted model to forecast sales for 1993.

2.1 – Theoretical Analysis

We will apply single, double, triple EWMA's to predict the sales in 1993. We will first use the 72 transformed data points (first 6 years) for model training using the Holt-Winters prediction function in R.

The **additive** Holt-Winters method is a popular technique used for forecasting time series data that has both trend and seasonality, like our data set. It has 3 main components, the level L_t , trend T_t and seasonality S_t . The method is detailed in the table above. The method updates these components at each time step and uses them to make forecasts.

The **multiplicative** Holt-Winters prediction formula is appropriate when the seasonal variations are proportional to the level of the time series. The formula has 3 main equations for updating the level L_t , trend T_t , and seasonality S_t components as well as one equation for making predictions. The method is detailed in **Table 1** above.

2.2 – Experimental Results

Single EWMA

For the single EWMA, we set the smoothing parameters β (trend) and γ (seasonal) to FALSE.

```
Holt-Winters exponential smoothing without trend and without seasonal component.

Call:
HoltWinters(x = MD, beta = FALSE, gamma = FALSE)

Smoothing parameters:
alpha: 0.3716002
beta : FALSE
gamma: FALSE

Coefficients:
[,1]
a 10.57251
```

Figure 2: Single EWMA Computer Printout

For the double EWMA, set γ (seasonal) to FALSE.

Double EWMA

```
Holt-Winters exponential smoothing with trend and without seasonal component.

Call:
HoltWinters(x = MD, gamma = FALSE)

Smoothing parameters:
alpha: 0.4219034
beta : 0.1566127
gamma: FALSE

Coefficients:
[,1]
a 10.7551734
b 0.1605617
```

Figure 3: Double EWMA Computer Printout

For the triple EWMA, all of the parameters are used.

Triple EWMA

```
Holt-Winters exponential smoothing with trend and additive seasonal component.

Call:
HoltWinters(x = MD)

Smoothing parameters:
alpha: 0.4379168
beta : 0
gamma: 1

Coefficients:
[,1]
a 10.139495294
b 0.029963187
s1 -0.646470508
s2 -0.416049411
s3 0.005888228
s4 -0.228125707
s5 -0.363832547
s6 -0.101330406
s7 0.017796278
s8 0.051425912
s9 0.038028697
s10 0.009006247
s11 0.386200208
s12 1.159267545
```

Figure 4: Triple EWMA Computer Printout

Interpretations of Computer Printouts

1. Single EWMA

- The smoothing parameter alpha is small, indicating that the current value is a weighted average of many past observations and the model gives more weight to historical data. The small alpha also implies that the model is smoothing out short-term fluctuations and emphasizing the long-term trend.

2. Double EWMA

- The larger alpha compared to the single EWMA suggests that the model is giving more weight to the most recent observations (the model reacts more quickly to recent changes.) The trend parameter beta is small, indicating that the slope of the linear trend is not significant. The model doesn't see a strong increase or decreasing trend over time.

3. Triple EWMA (Holt-Winters)

- The larger alpha indicates that the model relies heavily on the most recent data (past data has less influence. The beta of 0 suggests that there is no linear trend in the data. The gamma of 1 means that the seasonality pattern relies heavily on the recent cycle, showing that the seasonal component is based primarily on the most recent yearly cycle.

To see which model is the best, we can look at the prediction plot in **Figure 5**. From the figure above, the triple EWMA model has the best performance amongst the 3 models as the data and the prediction are very similar. It considers both the trend and seasonal components, which align with real-world scenarios. The triple EWMA model also has the smallest SSE as seen in **Table 2**, which indicates a good fit.

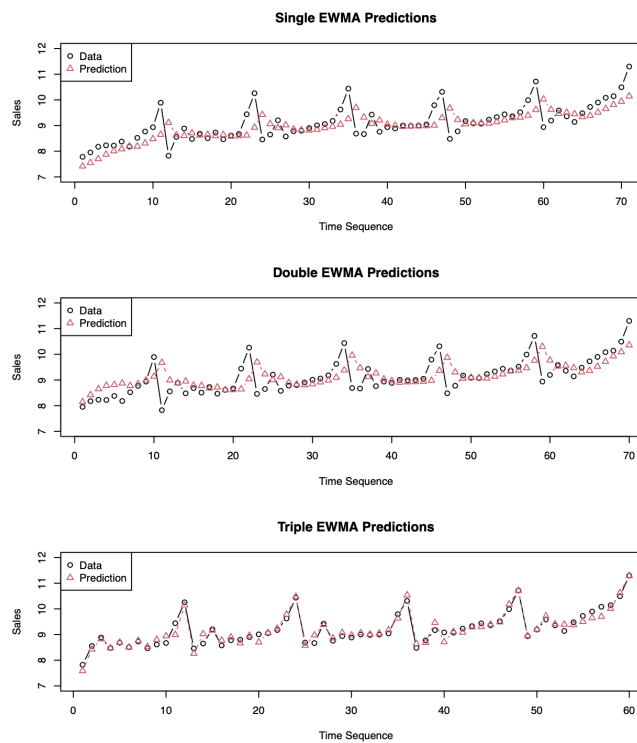


Figure 5: Predictions for all 3 EWMA

SSE of the 3 models:

Model	Single EWMA	Double EWMA	Triple EWMA
SSE	20.894080	23.329033	1.764109

Table 2: Sum of Squared Errors for Single, Double, Triple EWMA

2.3 – The Four Metrics Analysis

From the R outputs, we determined the best model is the triple EWMA. In this model, $\beta = 0$, showing that the trend effect stays the same throughout the time sequence. There is a high $\gamma = 1$ value, meaning that the estimation of the seasonality at each time point is heavily related to the data points that are close to it. We have four metrics: mean forecast error (MFE), mean absolute deviation (MAD), mean absolute percentage error (MAPE), mean squared error (MSE).

$$MFE = \frac{\sum Y_t - \widehat{Y}_t}{n} \quad MAD = \frac{\sum |Y_t - \widehat{Y}_t|}{n} \quad MAPE = \frac{\sum \left| \frac{Y_t - \widehat{Y}_t}{Y_t} \right|}{n} \quad MSE = \frac{\sum (Y_t - \widehat{Y}_t)^2}{n}$$

The closer the MFE value is to 0, the better. For MAD, MAPE, MSE, the smaller the value the better. I will explain more about what the metrics reveal in a later section.

Model Metric	Single EWMA	Double EWMA	Triple EWMA
MFE	0.1195834	-0.0441414	-0.002615137
MAD	0.3971345	0.4124712	0.132432909
MAPE	4.3012736	4.5313242	1.446937116
MSE	0.2942828	0.3332719	0.029401823

Table 3: MFE, MAD, MAPE, MSE values for each EWMA model

The triple EWMA model has an MFE value closest to 0 and the smallest MAD, MAPE, MSE values. Based on these values, we can reconfirm that the triple EWMA model is the best model for predicting results.

3 – Model Forecasting

3.1 – Introduction

For this section, we use R to print out the forecasting models. The triple EWMA model outperformed the single and double EWMA models, which is logical as it accounts for seasonality. The data has a clear trend and seasonal peaks; therefore, the triple EWMA model will align better than the other 2 models.

3.2 – Comparison

The forecasting reports in **Figure 6** show the forecast for upper and lower bounds. For the single EWMA and double EWMA, the difference between them becomes larger as the time sequence furthers, showing that the forecasting errors also increase as time increases. The difference between the lower and upper bound is close to consistent in the triple EWMA model, showing that the forecasting error is minimized compared to the single/double EWMA models. The upper and lower bounds are also closer together in the triple EWMA model than in the other models.

In the detailed comparison shown in **Figure 7**, the single EWMA has a constant forecast with high error variability, indicating poor forecasting. The double EWMA shows varying forecasts with consistently larger negative errors, indicating systematic overestimation. The triple EWMA has the most accurate forecasts with smaller, mostly negative errors, indicating the best fit among the three sets.

In the forecasting result plot **Figure 8**, we can see the data points of the forecast value and actual data values, noted as errors. The forecast data points are fairly similar to the actual data values in the triple EWMA (the error for triple EWMAs is small). The single EWMA is the next most similar, then the double EWMA.

		fit	upr	lwr
Jan 1993	10.57251	11.61697	9.528043	
Feb 1993	10.57251	11.68675	9.458261	
Mar 1993	10.57251	11.75242	9.392599	
Apr 1993	10.57251	11.81461	9.330403	
May 1993	10.57251	11.87384	9.271176	
Jun 1993	10.57251	11.93049	9.214530	
Jul 1993	10.57251	11.98486	9.160154	
Aug 1993	10.57251	12.03722	9.107795	
Sep 1993	10.57251	12.08777	9.057244	
Oct 1993	10.57251	12.13669	9.008326	
Nov 1993	10.57251	12.18412	8.960893	
Dec 1993	10.57251	12.23020	8.914816	

	fit	upr	lwr
Jan 1993	10.91574	12.05205	9.779420
Feb 1993	11.07630	12.34069	9.811908
Mar 1993	11.23686	12.64932	9.824396
Apr 1993	11.39742	12.97590	9.818940
May 1993	11.55798	13.31855	9.797414
Jun 1993	11.71854	13.67567	9.761421
Jul 1993	11.87911	14.04592	9.712296
Aug 1993	12.03967	14.42820	9.651138
Sep 1993	12.20023	14.82161	9.578849
Oct 1993	12.36079	15.22541	9.496175
Nov 1993	12.52135	15.63897	9.403739
Dec 1993	12.68191	16.06176	9.302065

	fit	upr	lwr
Jan 1993	9.522988	9.861859	9.184117
Feb 1993	9.783372	10.153312	9.413433
Mar 1993	10.235273	10.633867	9.836679
Apr 1993	10.031222	10.456544	9.605900
May 1993	9.925479	10.375946	9.475012
Jun 1993	10.217944	10.692225	9.743663
Jul 1993	10.367034	10.863989	9.870079
Aug 1993	10.430627	10.949265	9.911988
Sep 1993	10.447193	10.986644	9.907742
Oct 1993	10.448133	11.007624	9.888643
Nov 1993	10.855291	11.434126	10.276455
Dec 1993	11.658321	12.255877	11.060765

Forecast Results (single)

Forecast Results (double)

Forecast Results (triple)

Figure 6: Forecast Results for Single, Double, Triple EWMA model

	Actual	Forecast	Error
Jan 1993	9.234373	10.57251	-1.3381346
Feb 1993	9.329623	10.57251	-1.2428852
Mar 1993	9.990896	10.57251	-0.5816122
Apr 1993	9.761770	10.57251	-0.8107377
May 1993	9.680206	10.57251	-0.8923020
Jun 1993	9.830999	10.57251	-0.7415088
Jul 1993	10.171801	10.57251	-0.4007065
Aug 1993	10.260691	10.57251	-0.3118173
Sep 1993	10.325659	10.57251	-0.2468486
Oct 1993	10.335962	10.57251	-0.2365456
Nov 1993	10.750093	10.57251	0.1775854
Dec 1993	11.558479	10.57251	0.9859708

	Actual	Forecast	Error
Jan 1993	9.234373	10.91574	-1.681362
Feb 1993	9.329623	11.07630	-1.746674
Mar 1993	9.990896	11.23686	-1.245963
Apr 1993	9.761770	11.39742	-1.635650
May 1993	9.680206	11.55798	-1.877776
Jun 1993	9.830999	11.71854	-1.887545
Jul 1993	10.171801	11.87911	-1.707304
Aug 1993	10.260691	12.03967	-1.778977
Sep 1993	10.325659	12.20023	-1.874570
Oct 1993	10.335962	12.36079	-2.024829
Nov 1993	10.750093	12.52135	-1.771259
Dec 1993	11.558479	12.68191	-1.123436

	Actual	Forecast	Error
Jan 1993	9.234373	9.522988	-0.28861472
Feb 1993	9.329623	9.783372	-0.45374953
Mar 1993	9.990896	10.235273	-0.24437740
Apr 1993	9.761770	10.031222	-0.26945216
May 1993	9.680206	9.925479	-0.24527282
Jun 1993	9.830999	10.217944	-0.38694490
Jul 1993	10.171801	10.367034	-0.19523249
Aug 1993	10.260691	10.430627	-0.16993615
Sep 1993	10.325659	10.447193	-0.12153335
Oct 1993	10.335962	10.448133	-0.11217115
Nov 1993	10.750093	10.855291	-0.10519725
Dec 1993	11.558479	11.658321	-0.09984241

Error Results (single)

Error Results (double)

Error Results (triple)

Figure 7: Detailed Comparison among Single, Double, Triple EWMA Models

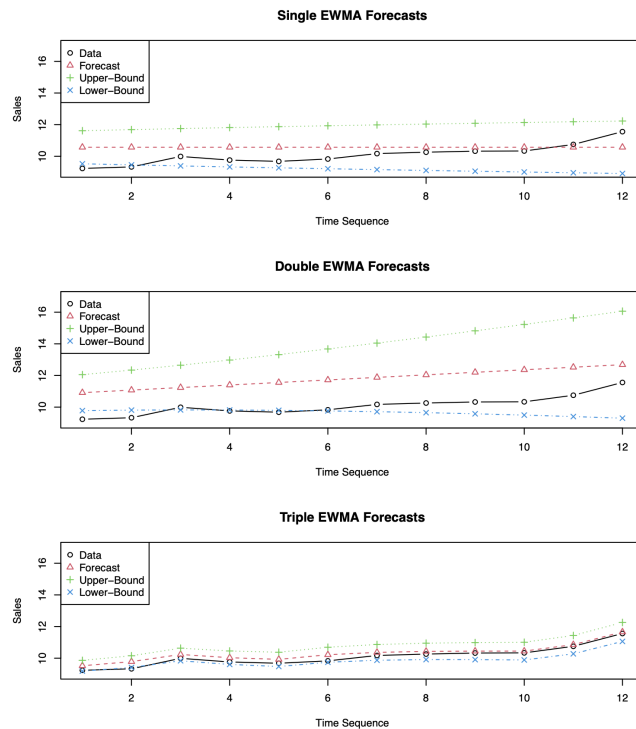


Figure 8: Forecasting Results for Single, Double, Triple EWMA Model

Looking at the four metrics again in **Table 4** (MFE, MAD, MAPE, MSE), we can see that the triple EWMA is the best model out of the 3 models. The MAD, MAPE, and MSE values are the smallest for the triple model, and the MFE value is the closest to zero out of all the models. Because all of the MFE values are negative, it indicates that all EWMA models are overestimating future values.

The single EWMA model has better metrics compared to the double EWMA model, suggesting it performs better than the double EWMA model. The metrics in order of best to worst are: triple, single, and then double.

	TDsMetrics	TDdMetrics	TDtMetrics
MFE	-0.4699619	-1.696279	-0.22436036
MAD	0.6638879	1.696279	0.22436036
MAPE	6.7116250	16.901458	2.27876189
MSE	0.5879444	2.940629	0.06223156

Table 4: Metrics for the Forecasting Results

4 – Comparing Results

In **Table 5**, we compared the four metrics (MFE, MAD, MAPE, MSE) of the predictions and the forecasting models. From this table, we can discern that the prediction model is better at prediction than the forecasting model. This is evident because the MFE values for the single, double, and triple EWMA models are all closer to 0 in the prediction model than in the forecasting model. Additionally, the prediction model has smaller values of MAD, MAPE, and MSE compared to the forecasting model.

The reason for this is that the data used to construct the three EWMA models are the same set of data used in the prediction case. This means the model is tested on the data it was trained on, leading to better performance metrics. In contrast, in the forecasting case, the model results are compared with new testing data, which the model has not seen before. This typically results in higher error metrics because the model's performance is evaluated on its ability to generalize to unseen data.

	TDsMetrics	TDdMetrics	TDtMetrics			MDsMetrics	MDdMetrics	MDtMetrics
MFE	-0.4699619	-1.696279	-0.22436036		MFE	0.1195834	-0.0441414	-0.002615137
MAD	0.6638879	1.696279	0.22436036		MAD	0.3971345	0.4124712	0.132432909
MAPE	6.7116250	16.901458	2.27876189		MAPE	4.3012736	4.5313242	1.446937116
MSE	0.5879444	2.940629	0.06223156		MSE	0.2942828	0.3332719	0.029401823

Table 5: Combined Prediction (right) and Forecasting (left) Metrics for Single, Double, Triple EWMA Models

The combined prediction forecasting plot below shows that the fitted value is closer to the actual value in the prediction case. But, the fitted value and actual value are not nearly as close in the forecasting case.

The single EWMA captures the overall trend of the sales data but has some lag; its forecast shows a simple extrapolation of the recent trend, indicating a potential sharp rise in sales. This model overshoots and does not account for seasonality or trend changes.

The double EWMA model provides a slightly better fit to the data, capturing the trend more accurately. The forecast indicates a more moderate increase in sales compared to the single model. It suggests consideration of trend but doesn't fully account for seasonality.

The triple EWMA model fits the data the best, capturing both the trend and seasonality effectively, showing minimal lag and a good responsiveness to fluctuations. Its forecast is the most nuanced, showing a rise that is more aligned with the observed seasonal patterns. This is because this model includes level, trend and seasonality.

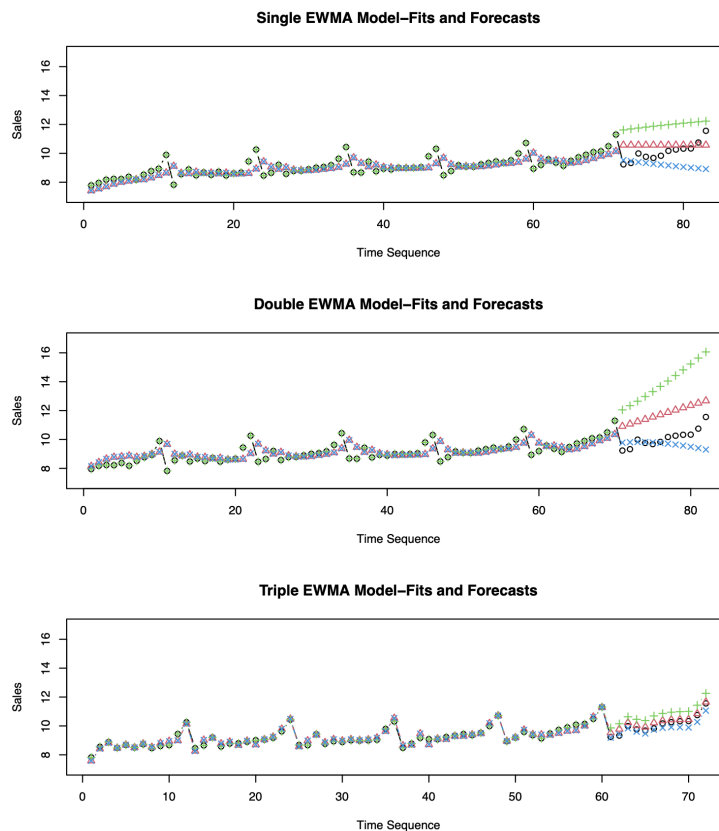


Figure 9: Combined Prediction Forecasting Results for the 3 Models

In conclusion, the better metric is the prediction case over the forecasting case. This is proven by the four metrics (MSE, MAD, MAPE, MSE) and the combined plot as seen in **Figure 9**.

